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The Relationships Between Reaction Time Scores and False Start Penalties of Offensive Linemen in the National Football League (NFL)

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Abstract

Each year in the United States, many factors are considered in the National Football League's draft process, including scouting reports, interviews, medical tests, and measures of players' strength, speed, anthropometry, and personality. Recently, an increased emphasis has been placed on players' psychometrically measured cognitive abilities. Historically, offensive linemen have been excluded from NFL research, since they do not have individual performance statistics. In this study, we attempted to fill that research gap by investigating correlates of offensive linemen's measured vigilance/impulsivity. We obtained archival player data from 85 offensive linemen in the years from 2014–2021 prior to each of these players' NFL draft. We then used false start penalties per game (FSPG) as an index of their vigilance/impulsivity (and general success) on the field, and we related those values to Reaction Time tests through hierarchical multiple regression analyses. After controlling for draft placement, these players' higher accuracy on a Choice Reaction Time task accounted for a statistically

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significant percentage of the variance associated with their FSPG statistics. We discuss the practical utility for player development of these and other cognitive assessments now used in the NFL.

Keywords
cognition in sport, american football, nfl draft, reaction time

Introduction

At the start of the 2023 American football season, the aggregate value of the National Football League (NFL) was estimated at $163 billion, with the value of the average team placed at $5.1 billion (Ozanian & Teitelbaum, 2023). In the context of these major economic pressures, all 32 NFL teams go to great lengths each year to compete for a championship, including each team's heavy investment in predicting and improving individual player's athletic performance.

The annual talent identification process for players entering the NFL draft includes a review of the players' anthropometric measurements and physical attributes (e.g., statistics related to their size, strength, and speed) and measures of their psychological functioning, including both their cognitive abilities and personality functioning. The development, measurement quality, and practical utility of both these physical and psychological measures have important limitations. For instance, research to date has shown inconsistent relationships between player performance on isolated physical tests and their subsequent NFL game performance (Lyons et al., 2011). Similarly, personality assessments have limitations, including a self-report bias that can contribute to inaccurate personality assessment (Roth et al., 1986). Additionally, since longitudinal research has shown that changes in personality tend to correlate with changes in life satisfaction, work environment, and socioeconomic status over time (Boyce et al., 2012; Le et al., 2014), an athlete whose personal wealth is changed dramatically by a multi-million-dollar NFL contract can experience major changes from initial personality findings.

Perhaps the domain with the greatest research advancements in recent years is cognitive functioning. Considerable research over 40+ years has shown relationships between cognitive functioning and athletic performance (Farrow et al., 1998; Faubert, 2013; Vaeyens et al., 2008). For instance, Jacobson and Matthaeus (2014) found that athletes scored significantly higher on measures of inhibition and problem-solving than non-athletes, and, in a meta-analysis conducted by Voss et al. (2010), athletes scored higher than non-athletes on measures of processing speed and varied cognitive attention tasks. Some researchers and theorists have argued that the cognitive domain may be the most important distinguishing factor between elite and non-elite athletes (Zaichkowsky & Peterson, 2018). Yet, for decades, the Wonderlic Personnel Test was the sole NFL measure of intellectual ability (Wonderlic, 2013), even while its scores

have not shown predictive validity with respect to NFL player performance (Kuzmits & Adams, 2008; Lyons et al., 2009). This seemingly led to the NFL's 2022 announcement that the Wonderlic Personnel Test would no longer be administered to all prospective players at the NFL Scouting Combine (O'Connell, 2022).

Among new cognitive tests developed for use with prospective NFL players, the Player Assessment Test (PAT; Siena Consulting, 2013) was designed to measure learning styles, decision-making skills, core intellect, and the ability to respond to unexpected stimuli (Breer, 2013; Howard, 2013). Unfortunately, despite the 10-year history of the PAT, there are no peer-reviewed research studies that have identified significant relationships between scores on factors of the PAT and aspects of NFL game performance. Other, newer cognitive measures (e.g., APTUS, S2) also lack published empirical support for NFL applications.

The one cognitive assessment tool with demonstrated significant relationships between test scores and NFL performance is the Athletic Intelligence Quotient (AIQ; Bowman et al., 2020; Boone, 2024). The AIQ, based on the most empirically supported theory of intelligence, the Cattell Horn Carroll (CHC) Theory (Schneider & McGrew, 2018) was first introduced at the NFL Scouting Combine in 2012. Bowman et al. (2020) revealed statistically significant correlations and multiple regression predictions between the AIQ's four factor scores and NFL metrics (e.g., Weighted Career Approximate Value (CAV) and Rushing Yards per game). Further, in a current study, NFL quarterbacks with higher AIQ scores threw for more touchdowns and passing yards and had higher quarterback ratings (QBR) and CAV (Boone, 2024). The AIQ has also shown utility in differentiating players in the National Basketball Association (NBA) from those in leagues that are presumed to include less skilled players (e.g., the G League or International Leagues) (Hogan et al., 2023), with significant correlations between AIQ factor scores and specific basketball game statistics (e.g., Player Efficiency Rating (PER), Effective Field Goal Percentage (eFG), and Turnovers). AIQ scores have also been significantly related to performance statistics in professional baseball (Bowman et al., 2021).

In the initial NFL research on the AIQ (Bowman et al., 2020), investigators analyzed relationships between AIQ scores and position-specific game data for offensive and defensive players, but did not include offensive linemen (viz., left tackles, left guards, centers, right guards, and right tackles) in those analyses. While offensive linemen may be less well known to fans than other NFL players, they are critical to a team's success, because they provide offensive players the necessary protection and time to execute winning plays (Krywaruczenko, 2019). In fact, added time for quarterback decision-making has been significantly correlated with pass completions (Alamar & Weinstein-Gould, 2008). In addition to pass-blocking, offensive linemen are engaged in essential run-blocking for critical yardage gains. Monson (2023) reported that Pro Football Focus ranked the offensive lines for the two 2023 Super Bowl teams (Philadelphia Eagles and Kansas City Chiefs) 1st and 3rd best in the league, respectively.

A challenge in quantifying game performance of offensive linemen is that objective statistics are minimal; thus the number of penalties committed is a rare quantifiable

performance statistic for this position group. According to NFLpenalties.com, the most frequent penalties committed by offensive linemen in 2020 (45% of all their penalties) were false starts. The NFL Rule Book, Rule 7, Section 4, Article 2, defines a false start as "any quick abrupt movement by a single offensive player, or by several offensive players in unison, which simulates the start of the snap." The penalty for a false start is a loss of five yards from the line of scrimmage and a replay of the down. Clearly, these errors make it harder for an offensive team to obtain a first down, impacting the momentum of a drive and sometimes changing the course of a game. For example, Burke (2010) calculated that Alex Barron's 43 career false start penalties cost his team 24.4 potential points over the course of his career, or the equivalent of five points per season, leading in turn to a cost of 0.62 wins, or about 0.12 wins per season.

While an offensive line works as a single unit with interdependency in their play (Bradbury, 2008; Carroll, et al., 1988), if Player A incurs a false start in a game, it is safe to conclude that Player A is individually at fault for the infraction. Thus, this metric provides a limited but meaningful indication of a player's individual performance at a position that is difficult to evaluate. Although centers are critical members of the offensive line, they are not often called for false starts because they are the ones who hike the ball, rather than having to react immediately to someone else hiking it. Given the guards' closer proximity to the center and quarterback, it is logical that they would incur fewer false starts than tackles. In fact, according to NFLpenalties.com, eight of the most penalized offensive linemen for false starts in 2023 were tackles, while two were guards.

For the purposes of this study, false starts may be considered indications of impulsivity or loss of focus that may relate to cognitive abilities associated with vigilance and response inhibition. In previous research, self-reported levels of impulsivity have significantly predicted sport behaviors like unsportsmanlike conduct (Kang, Kim, & Lee, 2021). Moreover, in a study of handball players, demonstrated levels of impulsivity on neurocognitive assessments were positively correlated with technical faults and fouls (Lage et al., 2011). Such research provides a clear rationale for why offensive linemen who are less vigilant/more impulsive would likely commit more false starts per game than other players.

In the context of prior research, literature and logic, we investigated the relationships between neurocognitive tests of impulsivity and vigilance within the AIQ and NFL game-based false start penalties committed by offensive linemen. The AIQ has established reliability and has been empirically validated as a measure of sport specific cognitive abilities (Bowman et al., 2020, 2021; Hogan et al., 2023), but it has not been applied to offensive linemen to date. Although the AIQ is a broad cognitive measure, we focused on only two indices of specific relevance to false start penalties: Simple Reaction Time and Choice Reaction Time. These two measures on the AIQ provide both speed-based scores as well as separate accuracy scores, with the latter denoting a player's vigilance/impulsivity.

Hypotheses

First, based on patterns observed in the NFL, we hypothesized that, within our sample, tackles would have a higher rate of false start penalties per game than guards. Next, given the results of previous research on measures of abilities identified in CHC Theory, we hypothesized a relationship between Reaction Time Task accuracy measures and the number of false starts per game in the NFL. Additionally, we hypothesized that these two AIQ scores would explain a statistically significant increase, beyond draft pick and position, in the variance in false starts per game. Draft pick was chosen as the first independent variable covariate because the athlete's pick in the draft represents the player's estimated value based on the combination of multiple factors (e.g., strength, speed, size, scouting reports, personality, etc.). Player position was chosen as the second covariate, given the hypothesized pattern of false starts per game for offensive linemen. Ultimately, he AIQ should be able to improve predictive power beyond these variables if it is to provide utility. Thus, the following hypotheses were proposed:
H1: There will be a significant difference in false start penalties per game between tackles and guards.
H2: The two AIQ reaction time indices of vigilance and impulse control will be significantly correlated with false start penalties per game for offensive linemen. More specifically, the number of commission errors on the simple RT task will be positively related to false starts and accuracy on the Choice RT task will be negatively related to false starts.
H3: The two AIQ indices of vigilance and impulse control will provide a statistically significant increase in the explanation of variance in false starts per game, beyond draft pick and position.

Method

Participants

We administered the AIQ to 315 offensive linemen between 2014 and 2021, prior to their respective NFL Draft year. From these offensive linemen, we purposely excluded centers because, as hikers of the ball, they are far less susceptible to false start penalties. Among our 315 participants, 85 had played in at least 17 games by the end of the 2021 NFL season, which is equivalent to one full NFL season. This level of participation was considered the minimum threshold for inclusion in these analyses.

Ethical Considerations

Participation in the study was voluntary, and we obtained informed consent for this research from all participants in accordance with the NFL draft evaluation procedures. Additionally, only anonymized data were accessed and used for this study. Due to fact

that the data were anonymized and archival, this study was determined to have exempt status by an Institutional Review Board.

Assessment Instrument

Athletic Intelligence Quotient (AIQ). The Athletic Intelligence Test is a comprehensive measure of cognitive ability composed of 10 subtests. It is administered on a tablet by a software program. Subtests are presented in a fixed, successive order, with audio/visual instructions provided before the start of each task. The administration time for the AIQ generally ranges from 35–38 minutes. See Bowman et al. (2020) or Hogan et al. (2023) for additional information about each subtest and factor.

We analyzed results from two subtests on the AIQ. Both measures assess Reaction Time, which has been defined as the ability to react and/or make decisions quickly in response to simple stimuli (Flanagan et al., 2006). These abilities are typically measured by chronometric measures of reaction time. The AIQ includes a measure of Simple Reaction Time and Choice Reaction Time. Simple Reaction Time is the latency period (in milliseconds) of a response to a single stimulus (visual or auditory) presented at a particular point of time. By contrast, Choice Reaction Time requires the accurate and immediate processing of target stimuli among a series of two or more distractor stimuli, depending on which is signaled.

On the AIQ, both subtests yield scores based on the average reaction time for each trial, and separate scores that are derived based on the accuracy of the respondent. For the simple reaction time task, accuracy is best evaluated with number of commission errors or false alarms. For the choice reaction time task, the best measures is the *d* prime statistic, an indicator of sensitivity or discriminability. It is calculated based on the number of correct hits (pressing the button when the target appeared) and the number of commission errors (pressing the button prematurely, when a target has not appeared). Thus, these reaction time accuracy scores are not based on the speed of an individual's response but rather on their ability to remain vigilant and inhibit impulses.

Given the factors that likely affect a player's propensity to commit false start penalties, we postulated that their reaction time accuracy scores rather than speed-based (reaction time) scores or other cognitive ability scores would be most impactful. As offensive linemen get into position, they are essentially waiting for a signal. They must engage in the play as soon as possible after the ball is hiked. Reacting before the ball is hiked is a penalty; reacting late means your advantage in the play is lost. The most basic aspects of this play would appear to be like that of a Simple RT task. However, there are numerous distractors including the movements of the defense, movement in the backfield, fan noise, and a hard count from the quarterback, while the lineman must only attend to the correct cue, the action of the center. If they react to one of the distractors, they will incur a false start penalty. The cognitive requirements of this game situation appear related to the vigilance/impulse control required in the Choice RT Task.

Procedure

Trained administrators briefly described the assessment protocol before asking participants to provide informed consent, which included their express right to discontinue responding to assessment questions at any time. When the athletes arrived at the evaluation room, the administrator led them to the testing station and briefly explained the testing procedures. Next, the administrator initiated the computer program for the participants and presented them with headphones for audio instructions.

Data Analysis

We conducted all statistical analyses using the Statistical Package for the Social Sciences (SPSS 29.0; IBM Corp, Armonk, NY, USA), and we considered a p-value of $<$ .05 as statistically significant. Power analyses using the G*Power program (Faul et al., 2007) demonstrated that we had 80% power for detecting large effect sizes in all our planned analyses. We computed descriptive statistics (means and standard deviations) for all outcome measures. Additionally, we accounted for games played by calculating the number of False starts per Game (FSG), our primary dependent variable. For the AIQ measures, we focused on the commission errors for the simple RT and calculated d' for choice RT, as described by Macmillan and Creelman (1991; see also Stanislaw & Todorov 1999). We used a one-way analysis of variance (ANOVA) to test for differences between position (Guard, Tackle, and Both) on number of False Starts per Game, utilizing the REGWQ post hoc test. We used hierarchical regression to test the effects of draft pick[1], position, commission errors for simple RT, and *d'* for choice RT on number of False Starts per game. This same analysis also included the calculation of the zero-order correlations between False Starts per Game and the predictor variables.

Results

Our first hypothesis required a determination of whether the player's false starts in NFL games were related to their position on the field, with the expectation that there would be false start differences between players in tackle vs. guard positions. We ran a one-way between subjects ANOVA which demonstrated a significant difference between three groups (i.e., tackles, guards, and players who played both tackle and guard) F (2,84) = 3.96, p = .02. A Student-Newman-Keuls post-hoc test revealed that tackles (n = 36; M = .16, SD = .09) were more likely to have false starts than guards (n = 18, M = .10, SD = .05) or than players who played both positions (n = 31; M = .12, SD = 08).

Table 1 shows the descriptive statistics and the zero-order correlations addressing our second hypothesis that there would be a significant relationship between these RT accuracy scores and the player's FSPG average. The pattern of zero order correlations support the anticipated relationships between FSPG and both reaction time accuracy measures., The number of commission errors on the simple choice RT was positively

correlated with False starts per game; more errors meant more false starts. At the same time, *d'* for choice reaction time was negatively associated with FSPG; lower accuracy in this task led to more false starts.

We ran a hierarchical multiple regression on the data obtained from our NFL offensive linemen/participants, using their average FSPG as the dependent variable and the following predictor variables entered in this order: draft pick, position (guard, tackle, both), commissions for simple reaction time, and d' for choice reaction time. Additionally, given a significant main effect for position, we included the interaction terms involving position and reaction time measures from both the Simple Reaction Time and Choice Reaction Time tasks. Given that the interaction terms were not significant predictors, we dropped those terms and re-ran the analysis, with results shown in Table 2. This analysis revealed that, after controlling for draft pick and the main effect for position, the two reaction time accuracy measures contributed an additional 6.3% of the variance, ΔF (2, 76) = 3.00, p = .05; with the entire model explaining 21% of the variance, F (5, 76) = 3.91, p = .003. Specifically, the Choice Reaction Time score was a

Table 1. Descriptive Statistics and Zero-Order Correlations for FSPG, Draft Pick and Reaction Time Subtask Measures (N = 82).

	M	*SD*	Draft pick	Errors of commission simple reaction time	d' choice reaction time
Average FSPG	.13	.09	−.31**	.22*	−.29*
Draft pick			-	−.13	.16†
Errors of commission simple reaction time task	3.54	3.74		-	−.41***
d' choice reaction time task	3.56	.43			-

Note. †p < .10, *p < .05, **p < .01, ***p < .001.

Table 2. Hierarchical Regression of False Starts per Game as a Function of Draft Pick, Position, and Reaction Time Subtask Measures.

Step and predictor variables	R^2	ΔR^2	sr^2	β	SEB
Average FSPG					
Step 1	.09**	.09**			
Draft pick round			−.31**	−.31	.00
Step 2	.14**	.05			
Position			-	-	-
Step 3	.21**	.06*			
Errors of commission			.06	.06	.00
Simple RT					
d' choice RT			−.21*	−.23	.02

Note. †p < .10, *p < .05, **p < .01, ***p < .001.

unique predictor of average false starts per game, with lower accuracy (poorer vigilance and greater impulsivity) on the Choice Reaction Time task predicting more false starts. These findings are in line with our a priori hypothesis number 3.

Discussion

In recent years, mounting research has highlighted relationships among cognitive factors and performance in professional sports (Jacobson & Matthaeus, 2014; Voss et al., 2010). With physical traits and abilities and personality variables already being measured in granular detail, it makes sense that teams across all sports would begin to assess cognitive abilities in a similar way, to obtain incremental advantages in player selection and development. Beyond various tasks used in past cognitive research, the Athletic Intelligence Quotient (AIQ), appears to hold particular promise as a reliable and valid comprehensive cognitive assessment tool with implications across a range of professional sports (Bowman et al., 2020, 2021; Hogan et al., 2023).

In the current study, we applied a similar research methodology as prior investigators who correlated AIQ results with specific game performance, except that we studied an unheralded position group in American football, offensive linemen. Given the challenges of assessing the individual game performance of offensive linemen, we focused on the objective dependent variable of their number of false start penalties, which negatively impact the offensive team's goal of moving the ball into the opponent's end zone (Krywaruczenko, 2019).

In line with our first hypothesis, we found a significant difference among the number of false starts for players who played different positions on the offensive line. Specifically, tackles incurred significantly more false start penalties per game than guards or players who played both tackle and guard. This is consistent with the pattern of the most penalized offensive linemen in the NFL last year, and makes intuitive sense, as tackles are farther away from the center, and therefore, may be less able to discriminate cues for the ball being hiked. There may also be some parallel to research on Pieron's Law on perceptual decision-making, which indicates that mean response times decrease (and response accuracy increases) as stimulus intensity increases (van Maanen et al., 2021). Specifically, closer proximity to the source (i.e., center hiking the ball) may represent greater stimulus intensity.

Consistent with our second hypothesis, players' levels of vigilance and impulsivity were significantly correlated with the number of false start penalties they incurred per game. In the Simple Reaction Time Task, the greater number of commission errors was positively related to the greater number of FSPG. Additionally, as hypothesized, the measure of accuracy for the Choice Reaction Time task was significantly negatively correlated with FSPG and provided a statistically significant increase in explained variance in FSPG beyond draft pick and position (i.e., Hypothesis 3).

Interestingly, while this relationship emerged for Simple Reaction Time accuracy, it was not a unique predictor when this variable was entered into the regression model with Draft Pick, position, and d' for CRT. However, if an offensive lineman struggles

with impulsivity/vigilance during the more complex Choice Reaction Time test scenarios, they are at greater risk of committing false start penalties, and this is a unique contributor to the overall model, above and beyond the effect of draft pick and position. Again, this makes intuitive sense, given that the presence of multiple distracting game factors (e.g., hard count from the quarterback, movement from a defender suggesting a potential blitz) may affect a player's ability to react immediately, but not prematurely to the hiking of the ball.

Overall, our findings add to a growing body of research underscoring the importance of cognitive factors in sports and the utility of measuring specific cognitive abilities that have relevance for game performance. Although there are implications in terms of scouting and talent selection, player development and peak performance may be the most critical applications of knowledge gained from applying cognitive assessment in sports. If team representatives and players can understand the unique cognitive strengths needed from players in each position, teams may determine how to set players up for success by matching players' cognitive profiles with given positions, schemes, and strategies, and players may learn what skills are most important to develop. For instance, a wide receiver with weaknesses in visual spatial processing may do better with simpler route trees (e.g., "go route," etc.). Similarly, an offensive lineman who struggles with impulsivity may perform better when closer to the ball (i.e., playing guard) to be more in rhythm with the count and snap than if farther away (i.e., playing tackle).

Further, when teams and players understand their cognitive strengths and weaknesses, they can implement drills and strategies tailored to their needs. For example, an offensive lineman who struggles with vigilance/impulse control at the line of scrimmage, may need to spend extra time in practice working on his "get off" to various counts/cadences from the QB. Coaches may also incorporate noise or other forms of distraction to simulate some of the real challenges of game scenarios. Ultimately, by identifying these individual factors pre-draft, teams may be able to identify and address potential concerns earlier in the player development process. Conversely, if coaches and team personnel lack this kind of information ahead of time, they may experience significant frustration by having to acquire it more slowly.

Limitations and Directions for Further Research

With respect to limitations, the metric we used of false start penalties clearly does not fully assess the complexity of offensive line play. As such, we recommend that investigators identify additional factors, such as individual player rankings (e.g., Pro Football Focus) in future research. Future researchers should also allow for the inclusion of centers, since centers also play an important role on the offensive line. The inclusion of additional measures of vigilance/impulsivity or other cognitive abilities would also allow comparisons of which tasks are most sensitive to aspects of offensive line play in the NFL. American footbal has several distractors that can lead to increased RT or even false start penalties during the game. A CRT controlled situation is a

possibility that can lead to increased preparation time for the response and wrong choice, and the results can give some insights about distractors in the game. Further, researchers might obtain a clearer understanding of the variance in NFL false starts explained by cognitive factors by controlling for the number of false starts incurred during players' college football careers.

Even in terms of penalties, there are other common infractions on the offensive line, such as holding penalties, that may correlate significantly with cognitive factors. Additionally, to address the player development implications noted above, researchers could explore the efficacy of different coaching strategies on penalties incurred for offensive linemen who struggle with vigilance/impulse control. Follow-up research might also address the defensive side of football, perhaps involving a similar penalty to false starts, known as an encroachment, defined by when "a defensive player enters the neutral zone and contacts an offensive player or the ball prior to the snap, or interferes with the ball during the snap…" (National Football League, 2022).

Conclusion

In the end, though the unique contributions of measures of impulsivity and vigilance appear to be modest in the prediction of false start penalties for offensive linemen, our study provides further support for the role of cognitive assessment in professional sports and serves as an example of how research of this kind can fill important gaps in our understanding of optimal player characteristics. Practically, this information has immediate benefit for tailored coaching and player development strategies.

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Note

1. For the undrafted players in our sample, we entered a draft pick number of 260 which was two units above the cut-off for the drafted players.

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